Hallucination and Knowledge-Cutoff Probe for a Library Assistant

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Abstract

Large language models (LLMs) are increasingly deployed as conversational library assistants that answer questions about books, authors, availability and library policies. Although fluent, these systems have two weaknesses that limit their trustworthiness in a setting where accuracy matters. The first is hallucination, in which the model fabricates book titles, authors, ISBNs, call numbers, citations or service rules that do not exist. The second is the knowledge cutoff: a model knows nothing after its training date, so it cannot reliably answer questions about new acquisitions, current editions or recently revised policies, and it often fails to admit this limitation. This project develops a Hallucination and Knowledge-Cutoff Probe, a structured set of test queries built from a library's own catalogue, to measure how often and in what ways an assistant fails and how well it recognises the limits of its knowledge. The probe rests on a five-type hallucination taxonomy covering phantom items, metadata corruption, citation fabrication, policy invention and availability fabrication. Test queries are organised into three tiers: catalogue-grounded questions with objective answers, adversarial questions containing false premises, and unanswerable questions where the correct behaviour is to abstain or redirect the patron to a librarian. Together, these tiers separate errors caused by missing knowledge from errors caused by overconfidence, which require different remedies. A second probe family targets temporal knowledge, covering edition currency, newly published titles, policy drift and the model's awareness of its own cutoff. Responses are split into atomic claims and verified against catalogue records by independent annotators. Results are summarised using per-type hallucination rate, claim-level factual precision, temporal staleness rate, cutoff awareness score and deferral precision. The findings guide a mitigation layer that combines retrieval grounding over the catalogue, verbatim copying of identifiers, claim verification and an abstention policy, which is evaluated through ablation of each component. The expected outcome is a reproducible, low-cost evaluation procedure that lets a library compare candidate models, monitor an assistant after each model update, and deploy it safely, answering when it knows and deferring when it does not. The work also provides a reusable template that other libraries can adapt to their own collections, languages and service policies.

Keywords: hallucination, knowledge cutoff, library assistant, LLM evaluation, retrieval-augmented generation